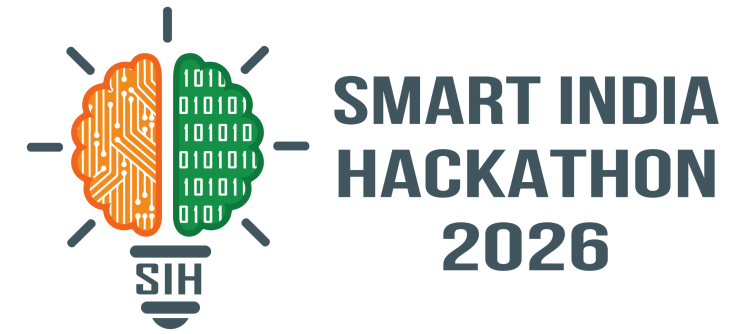
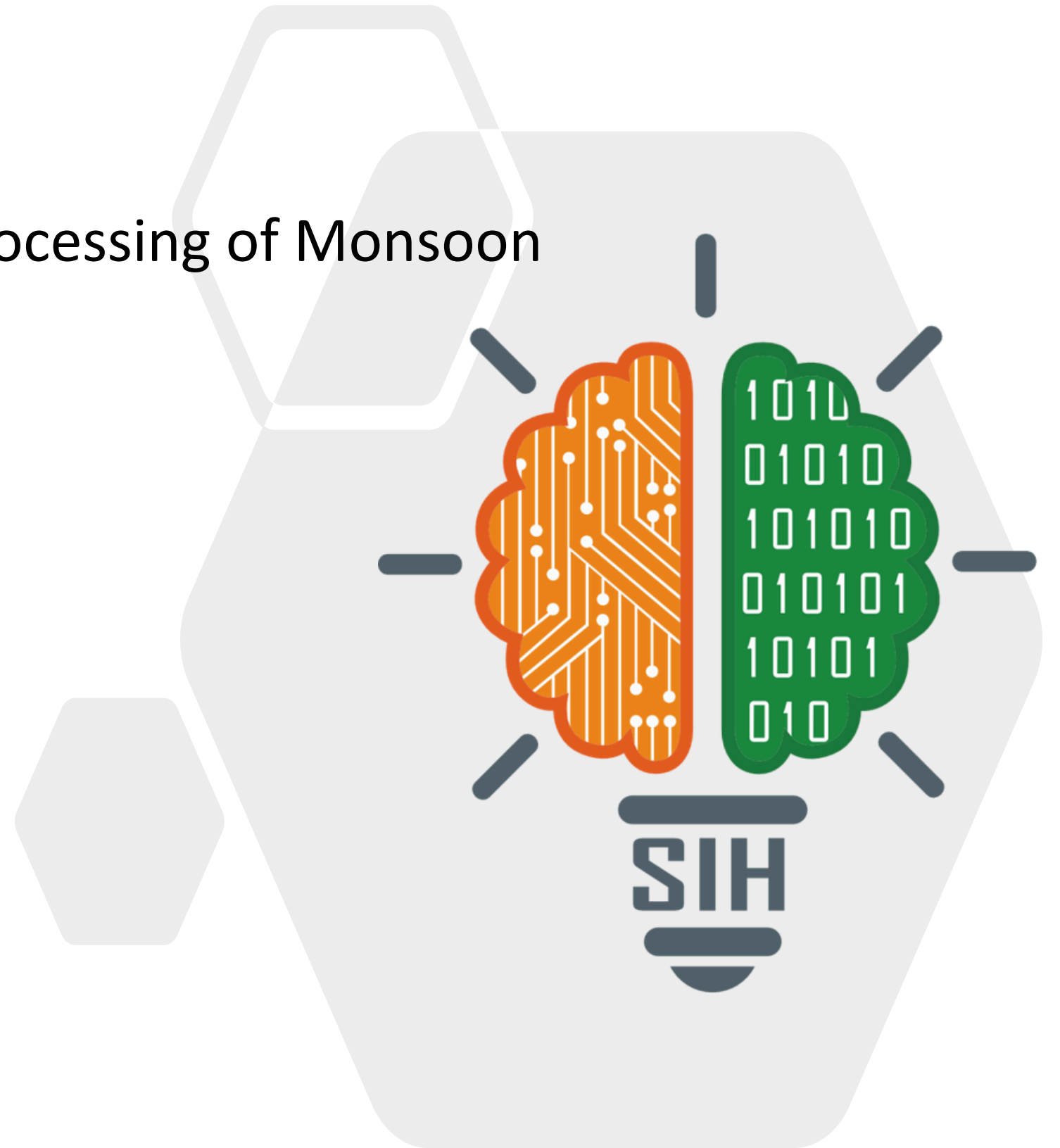
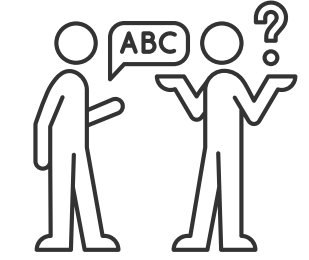
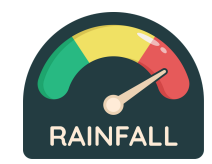


SMART INDIA HACKATHON 2026



- **Problem Statement ID** - SIH26080
- **Problem Statement Title** - Regime-Aware AI Post-Processing of Monsoon
- **Theme** - Miscellaneous
- **PS Category** - Software
- **Team ID** - 128223
- **Team Name** - BugHeBug





We turn **one-size-fits-all rainfall forecasts** into **regime aware predictions** transforming **raw NWP output** into **smarter, location specific forecasts** by learning how NWP errors change across monsoon conditions, then applying the right corrections to **detect and better heavy rainfall events**.

Demo Video: <https://youtu.be/vx3XgOpUlal?si=rHEJsEYdx0elkzdX>

Live Demo: <https://shreyashsri79--satquery.modal.run/>

REGIME AWARE FORECASTING

Instead of applying one correction to every forecast, the **system identifies** the **prevailing weather regime**: active/break monsoon, depression, normal; it then **selects the correction suited to that regime**

ERROR AWARE AI

The system goes beyond predefined weather regimes by **learning how the NWP model actually fails**
Wet bias → Missed extremes → displaced rainfall → under forecasting
The error regimes provide an additional data driven view of forecast failures

SOFT REGIME BLENDING:
Real weather can lie between regimes. Instead of forcing single label, the classifier produces regime probabilities and blends the corresponding corrections

REGIME CONDITIONED BIAS CORRECTION

Historical forecast and **observed rainfall** are used to learn correction for each: **Regime × Geography × Lead Day**
The system then used **Quantile Mapping + extreme-value modelling** to transform **raw NWP rainfall** into a **corrected forecast**.

HEAVY RAINFALL PROBABILITY

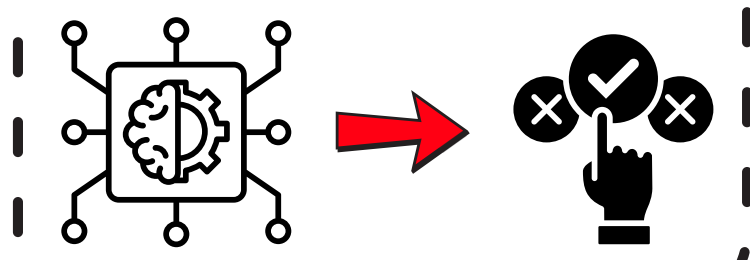
Along with a corrected rainfall value, the system predicts:
P(Heavy >= 64.5 mm/day)
P(Very Heavy >= 115.6 mm/day)
This converts the forecast into **actionable risk signal, rather than providing only single rainfall number.**

EXTREME RAINFALL CORRECTION: A dedicated extreme value tail model handles rainfall beyond the range captured by standard correction method

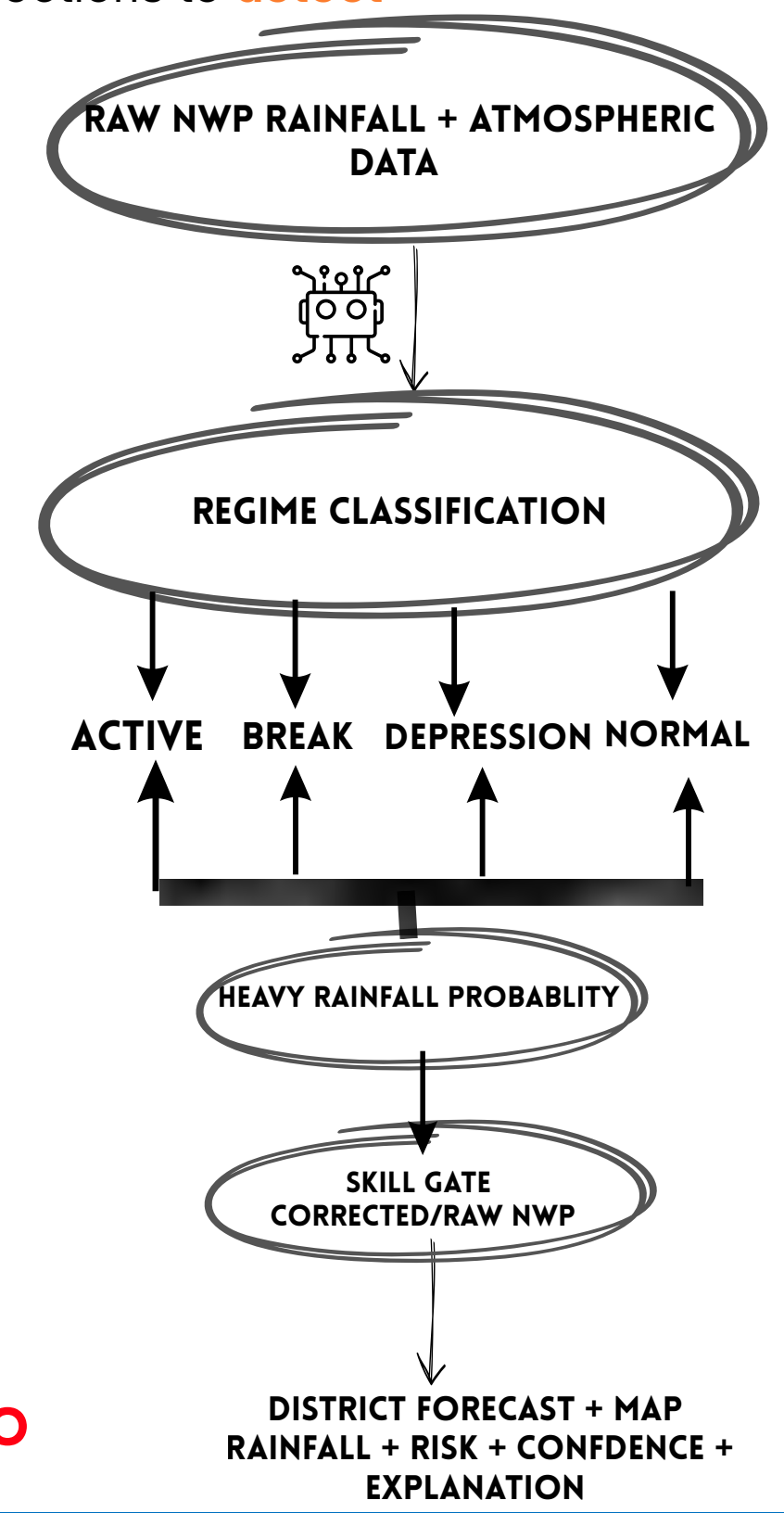
Verification first design
RAW NWP vs AI-corrected forecasts evaluated using RMSE, ETS, CSI, POD, FAR, FSS with leave one monsoon out validation across unseen monsoon seasons

SKILL GATED AI CORRECTION

The system continuously asks: **“Does our correction actually improve the forecast here ?”**
if historical validation shows the correction helps:
✓ **Use corrected forecast**
if it does not:
✗ **Fall back to the RAW NWP**
This **prevents the AI from blindly changing forecasts where correction is not reliable**

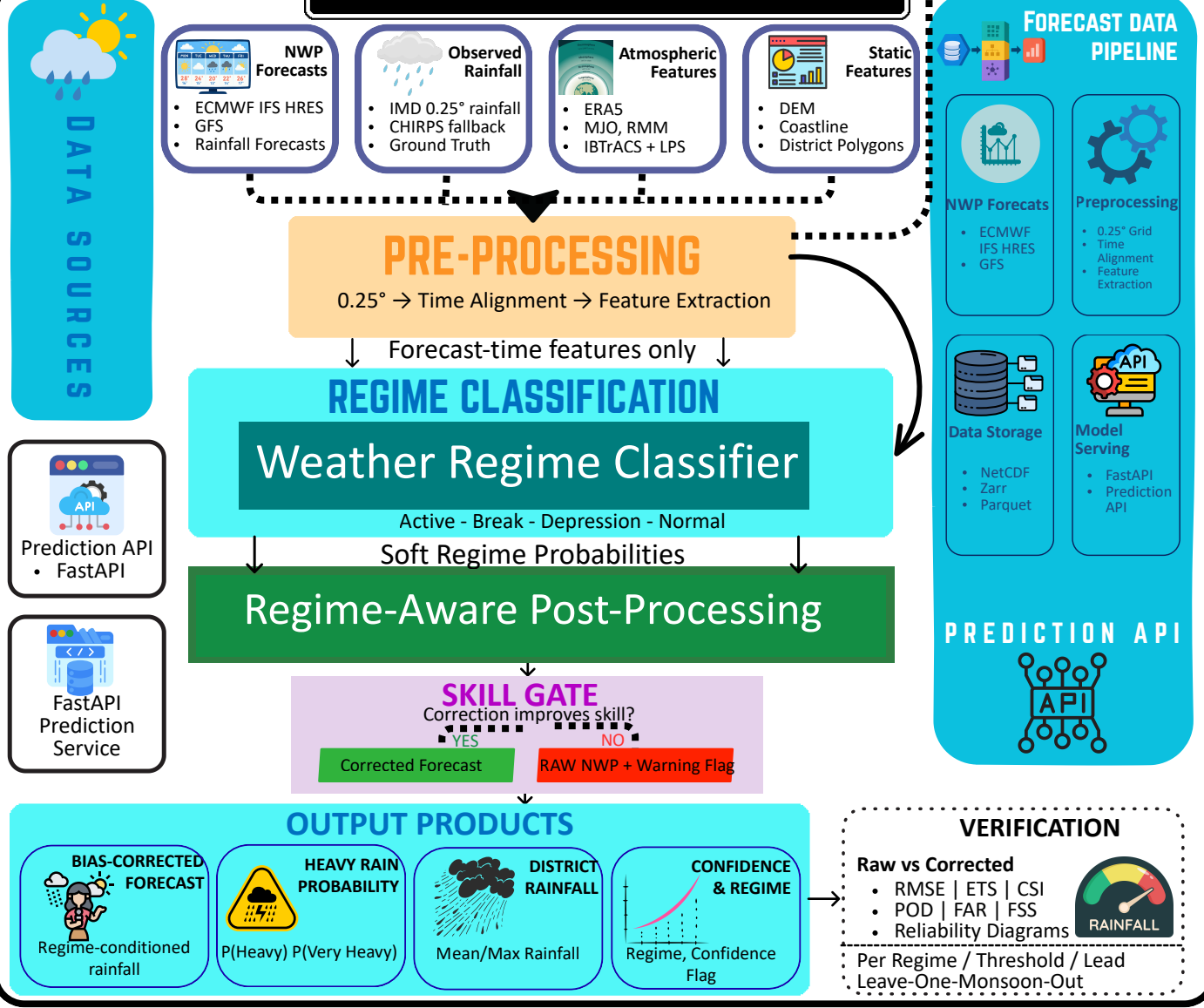


WE DON'T JUST CORRECT THE FORECAST WE LEARN WHICH KIND OF ERROR TO EXPECT, AND WHEN NOT TO CORRECT IT.

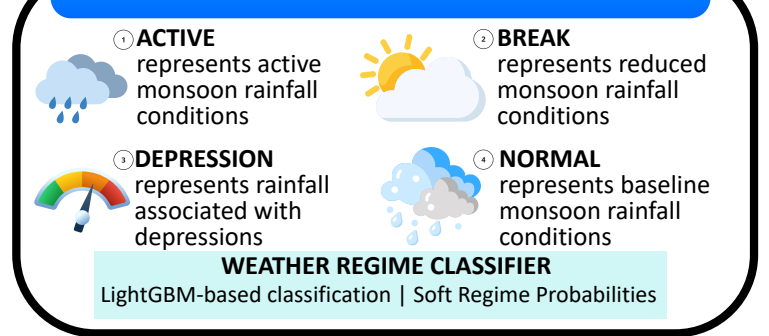


TECHNICAL APPROACH

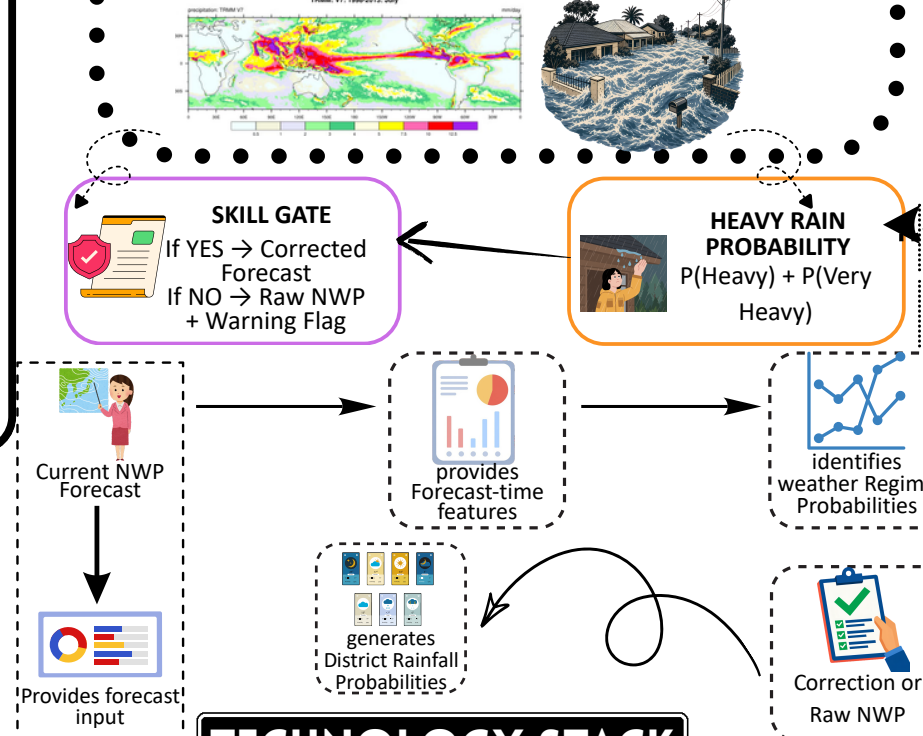
DATA & MODEL PIPELINE



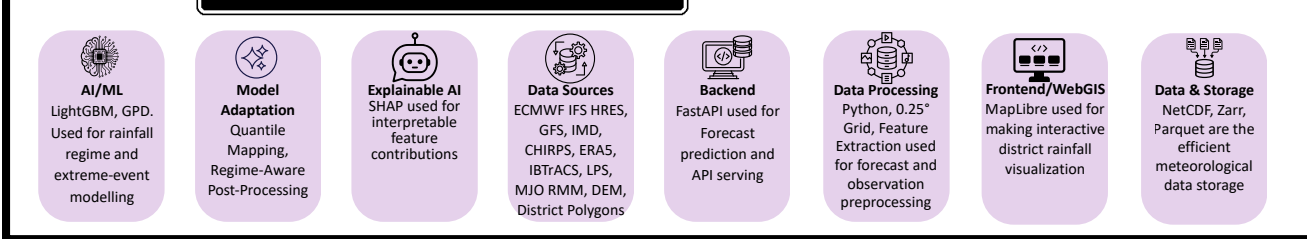
REGIME CLASSIFICATION



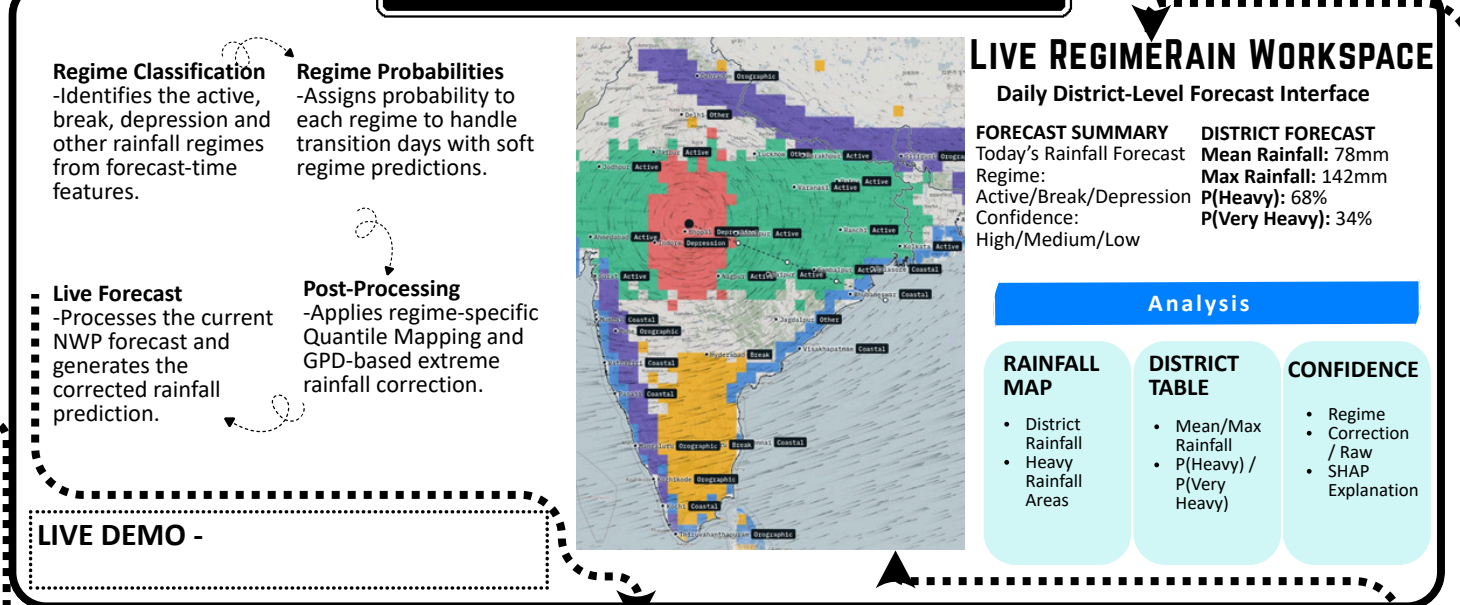
REGIME-AWARE POST-PROCESSING WORKSPACE



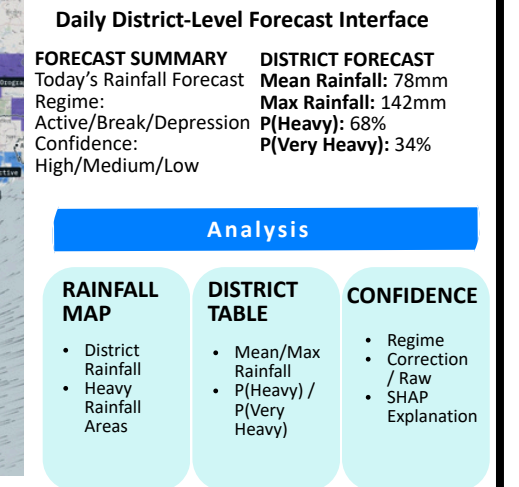
TECHNOLOGY STACK



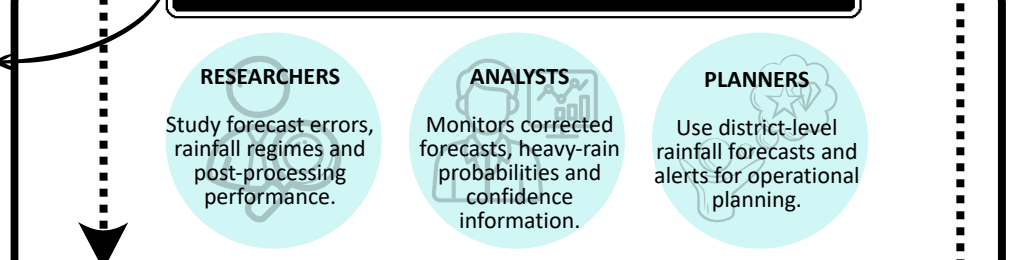
FORECAST ANALYST



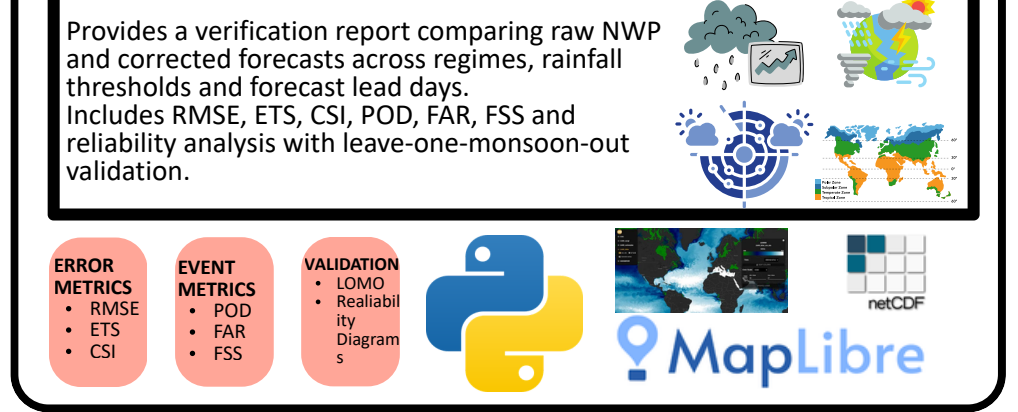
LIVE REGIMERAIN WORKSPACE



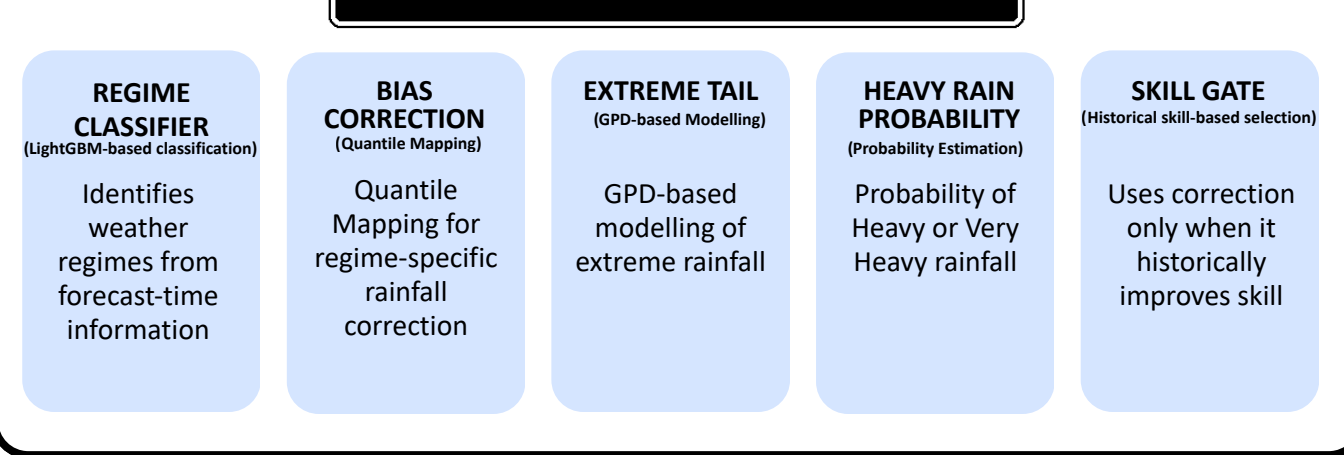
FORECAST USERS



RESULTS AND REPORTING



AI COMPONENTS

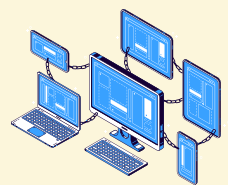


FEASIBILITY AND VIABILITY



Data Feasibility

Public weather inputs are available through **WeatherBench2**, **CHIRPS**, **NOAA GFS** and **ECMWF** open data, with no login required.



Computational Feasibility

The system operates on a **0.25° India grid** using lightweight tabular models and does not require GPU acceleration.



Technical Feasibility

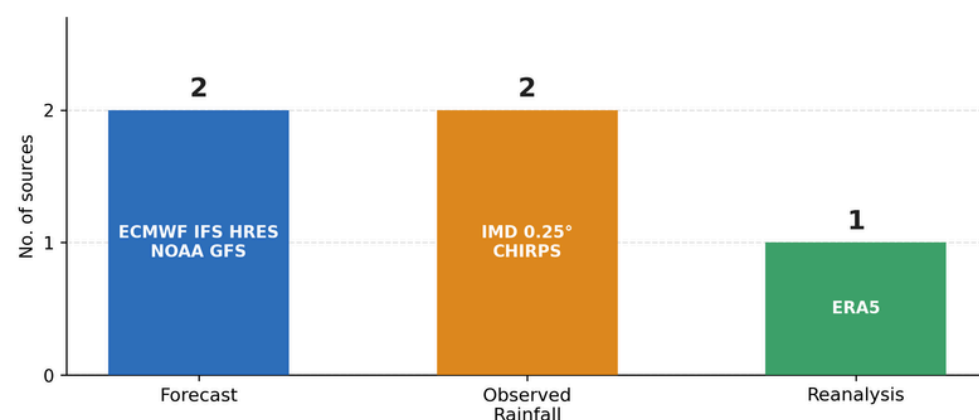
Python-based processing, adaptive weighting, regime detection, verification and a React/FastAPI dashboard support the proposed workflow.



Validation Feasibility

Performance can be evaluated using **RMSE**, **ACC**, **CSI**, **ETS**, **FSS** and **Brier metrics** with leave-one-year out verification and bootstrap confidence intervals.

CORE DATA SOURCES



Operational Viability

Supports daily forecast post-processing with regime-probability blending, skill-gated correction, and a district-level map and table.

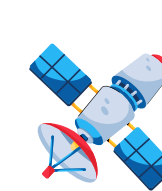
Integration Viability

The modular architecture can integrate NCUM/NEPS-G and future forecast sources into the same harmonized workflow without redesigning the core system.

Heterogeneous model coverage makes controlled model selection and temporal-overlap based evaluation essential for reliable blending.

CHALLENGES AND STRATEGIES

Integrating Forecast Sources and Maintaining Consistent Inputs



Model Availability Differences

- Forecast datasets may differ in archive period, resolution and availability.



Data Source Variability

- Forecast sources may differ in temporal coverage, resolution and availability.



Rainfall Truth Quality

- Rainfall verification needs reliable observational/gridded truth.



Forecast Training Models

- Live Forecast inputs must follow the same preprocessing and feature definitions used during training.

Ensuring Reliable Blending and Extreme-Event Forecasts



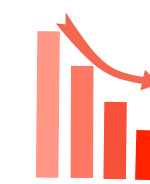
Extreme-Rainfall Handling

- Regime-aware correction preserves heavy and very-heavy rainfall behaviour.



Regime-Dependent Errors

- Model errors differ between active, break, depression and other weather regimes.



Small Regime Samples

- Tiny regime bins can produce noisy and unstable weight maps.



Evaluation Leakage

- Random train/test splits or using future information can produce misleading skill estimates.

Strategies to Overcome



Public-Data Adaptation

- Use standardized adapters to integrate NCUM/NEPS-G when data becomes available.



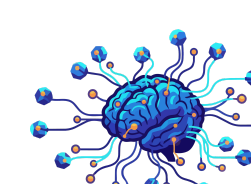
Consistent Model Inputs

- Standardize resolution, time alignment and feature definitions across available forecast sources.



Rigorous Validation

- Apply year/block-based testing, common masks and bootstrap confidence intervals.



Regime-Aware Correction

- Apply regime-specific bias correction using weather-regime probabilities.

IMPACT AND BENEFITS

Impacts



Trust Built In

Before going live, our system tests its correctness on past monsoon seasons for each district, regime and lead day. It checks whether the corrected forecast was actually closer to the real rainfall than the raw NWP forecast.



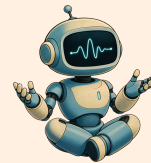
Actionable Output

Our system provides forecasters a ready-to-use district table and an interactive map showing mean and max rainfall, heavy-rain probability, weather regime and a confidence flag, enabling direct alert decisions.



Explainable Alerts

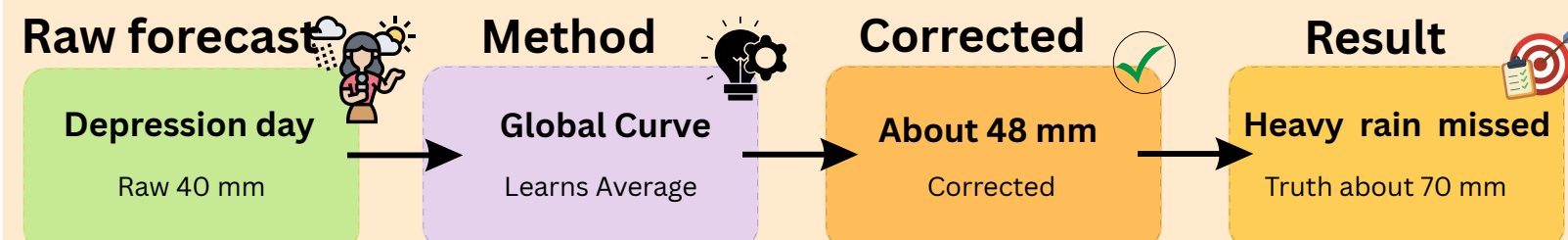
With every alert, a valid reason pops out so that the forecaster can understand the reason behind the alert. For example, depression regime $p=0.72$, strong low level jet, MJO phase 4.



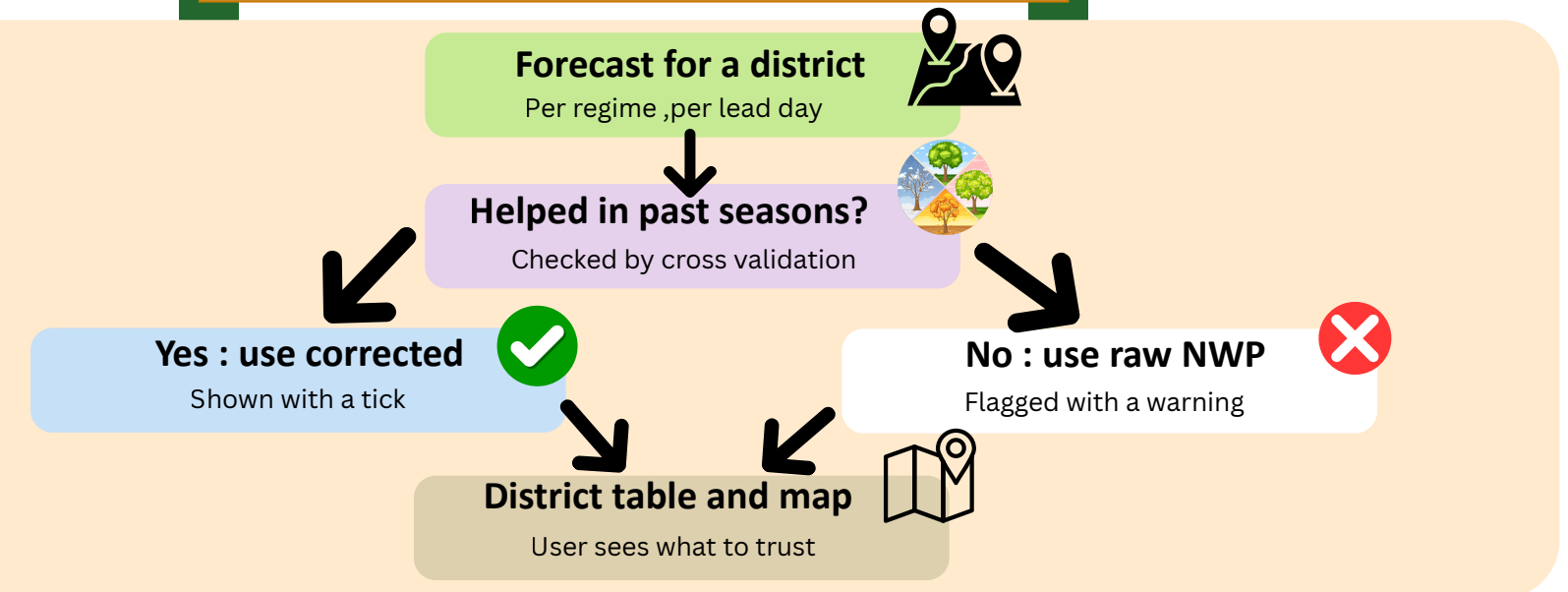
Diagnosing Model Errors for NCMRWF

We group days not only by weather type but also by how the model fails, and test which grouping corrects better. The result gives NCMRWF a diagnostic of where and why its model fails.

The Averaging Problem



Skill-Gated Correction with Raw NWP Fallback



Benefits

Social Benefits

Earlier, more trusted heavy-rain alerts protect lives in vulnerable regions, help people to get prepared for disaster and support timely farm advisories.

Economic Benefits

Gives accurate heavy-rain alerts that reduce flood and crop losses, support smart dam management and save forecaster effort, at low cost with no heavy GPU need.

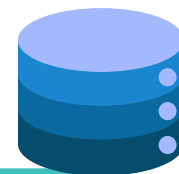
Scientific Benefits

Shows where and why the model fails, with honest repeatable testing. Helps scientists understand model errors and handle extreme rain better.

Institutional Benefits

Our system is not dependent on a single weather model. It works in transparency. NCMRWF can train our system on their NCUM/NEPS data and there is no need to learn new tools and formats.

Data & Thresholds



R1 -- NCMRWF. NCMRWF Unified Model and Data Assimilation System (NCUM writeup). National Centre for Medium Range Weather Forecasting, MoES.
nwp.ncmrwf.gov.in/ncmrwf/NCUM-Writeup.pdfR3

R3 -- NCMRWF. NCMRWF Ensemble Prediction System (NEPS): details.
www.ncmrwf.gov.in/NCMRWF-NEPS-Details.php

R4 -- NCMRWF. Data web portal (IMDAA reanalysis, NGFS forecasts).
www.ncmrwf.gov.in/data/

R5 -- MoES / INCOIS. Earth System Science Data Portal (ESSDP), NCMRWF holdings.
incois.gov.in/essdp/searchMetadata?...

R6 -- IMD Pune. High resolution $0.25^\circ \times 0.25^\circ$ daily gridded rainfall (NetCDF, 1901 onward). Data page. www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_NetCDF.html

R7 -- IMD Pune and NCMRWF. 0.25° merged satellite-gauge daily rainfall (NetCDF). www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_NetCDF_Merged.html

R8 -- India Meteorological Department, 2021: Standard Operation Procedure: Weather Forecasting and Warning Services. Ministry of Earth Sciences, New Delhi.
mausam.imd.gov.in/imd_latest/contents/pdf/forecasting_sop.pdf

R10 -- India Meteorological Department. District-wise rainfall distribution (rainfall statistics).
mausam.imd.gov.in/imd_latest/contents/rainfall_statistics_3.php

R36 -- Pai, D. S., M. Rajeevan, O. P. Sreejith, B. Mukhopadhyay, and N. S. Satbha, 2014: Development of a new high spatial resolution ($0.25^\circ \times 0.25^\circ$) long period (1901–2010) daily gridded rainfall data set over India and its comparison with existing data sets over the region. MAUSAM, 65(1), 1–18.
[doi:10.54302/mausam.v65i1.851](https://doi.org/10.54302/mausam.v65i1.851)

Regimes & Correction



R11 -- IITM Extended Range Prediction group. Active/Break spells selection criteria. Web page, Indian Institute of Tropical Meteorology, Pune. www.tropmet.res.in/erpas/files/active_break_selection.php

R12 -- Rajeevan, M., S. Gadgil, and J. Bhatle, 2010: Active and break spells of the Indian summer monsoon. *J. Earth Syst. Sci.*, 119(3), 229–247.
[doi:10.1007/s12040-010-0019-4](https://doi.org/10.1007/s12040-010-0019-4)

R13 -- Hurley, J. V., and W. R. Boos, 2015: A global climatology of monsoon low-pressure systems. *Quart. J. Roy. Meteor. Soc.*, 141(689), 1049–1064.
[doi:10.1002/qj.2447](https://doi.org/10.1002/qj.2447)

R14 -- Dimri, A. P., D. Niyogi, A. P. Barros, J. Ridley, U. C. Mohanty, T. Yasunari, and D. R. Sikka, 2015: Western Disturbances: A review. *Rev. Geophys.*, 53(2), 225–246.
[doi:10.1002/2014RG000460](https://doi.org/10.1002/2014RG000460)

R15 -- Phadtare, J. A., J. K. Fletcher, A. N. Ross, A. G. Turner, and R. K. H. Schiemann, 2022: Froude-number-based rainfall regimes over the Western Ghats mountains of India. *Quart. J. Roy. Meteor. Soc.*, 148(748), 3388–3405.
[doi:10.1002/qj.4367](https://doi.org/10.1002/qj.4367)

R16 -- Francis, P. A., and S. Gadgil, 2006: Intense rainfall events over the west coast of India. *Meteor. Atmos. Phys.*, 94(1–4), 27–42.
[doi:10.1007/s00703-005-0167-2](https://doi.org/10.1007/s00703-005-0167-2)

R17 -- Niranjana Kumar, K., M. S. Thota, R. Ashrit, A. K. Mitra, and M. N. Rajeevan, 2022: Quantile mapping bias correction methods to IMDAA reanalysis for calibrating NCMRWF unified model operational forecasts. *Hydrol. Sci. J.*, 67(6), 870–885. [doi:10.1080/02626667.2022.2049272](https://doi.org/10.1080/02626667.2022.2049272)

R18 -- Hewson, T. D., and F. M. Pilloso, 2021: A low-cost post-processing technique improves weather forecasts around the world. *Commun. Earth Environ.*, 2, 132.
[doi:10.1038/s43247-021-00185-9](https://doi.org/10.1038/s43247-021-00185-9)

R19 -- Zhang, T., Z. Liang, W. Li, J. Wang, Y. Hu, and B. Li, 2023: Statistical post-processing of precipitation forecasts using circulation classifications and spatiotemporal deep neural networks. *Hydrol. Earth Syst. Sci.*, 27(10), 1945–1960.
[doi:10.5194/hess-27-1945-2023](https://doi.org/10.5194/hess-27-1945-2023)

Probability & Verification



R21 -- Rasp, S., and S. Lerch, 2018: Neural networks for postprocessing ensemble weather forecasts. *Mon. Wea. Rev.*, 146(11), 3885–3900.
[doi:10.1175/MWR-D-18-0187.1](https://doi.org/10.1175/MWR-D-18-0187.1)

R26 -- Chakraborty, P., A. Sarkar, R. Bhatla, and R. Singh, 2021: Assessing the skill of NCMRWF global ensemble prediction system in predicting Indian summer monsoon during 2018. *Atmos. Res.*, 248, 105255.
[doi:10.1016/j.atmosres.2020.105255](https://doi.org/10.1016/j.atmosres.2020.105255)

R27 -- Shanker, G., A. Sarkar, A. Mamgain, R. Bhatla, and V. S. Prasad, 2024: Relative economic value of global ensemble prediction system of NCMRWF, India, for extreme weather events. *Quart. J. Roy. Meteor. Soc.*, 150(761), 1831–1846.
[doi:10.1002/qj.4674](https://doi.org/10.1002/qj.4674)

R31 -- WWRP/WGNE Joint Working Group on Forecast Verification Research. Forecast verification: issues, methods and FAQ. www.cawcr.gov.au/projects/verification/verif_web_page.html

R32 -- WWRP/WGNE Joint Working Group on Forecast Verification Research, 2009: Recommendations for the verification and intercomparison of QPFs and PQPFs from operational NWP models (Rev. 2, Oct 2008). WWRP 2009-1, WMO/TD-No. 1485. World Meteorological Organization, Geneva.
community.wmo.int/site/knowledge-hub/programmes-and-initiatives/world-weather-research-programme-wwrp/wwrp-publications

R34 -- Roberts, N. M., and H. W. Lean, 2008: Scale-selective verification of rainfall accumulations from high-resolution forecasts of convective events. *Mon. Wea. Rev.*, 136(1), 78–97.
[doi:10.1175/2007MWR2123.1](https://doi.org/10.1175/2007MWR2123.1)

R35 -- ECMWF, 2023: Scale-dependent verification of precipitation and cloudiness at ECMWF. ECMWF Newsletter, No. 174 (Winter 2022/23). www.ecmwf.int/en/newsletter/174/earth-system-science/scale-dependent-verification-precipitation-and-cloudiness

GITHUB LINK - <https://github.com/FrustratedBoy420/SIH>

LIVE DEMO - <https://shreyashsri79--satquery.modal.run/>

DEMO VIDEO - <https://youtu.be/vx3XgOpUlaI?si=rHEJsEYdx0elkzdX>